

Alex Lee

Waterloo, Canada | a55lee@uwaterloo.ca | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

EDUCATION

University of Waterloo

Sep. 2023 - Apr. 2028

Bachelor's in Data Science

Waterloo, ON

Relevant Courses: Data Structures, Algorithms, OOP, Statistics, Linear Models, Linear Algebra, Calculus

SKILLS

Languages: Python, C/C++, SQL(MySQL, PostgreSQL, MongoDB), R, Java, Bash

Libraries: Scikit-learn, TensorFlow, PyTorch, Pandas, NumPy, Matplotlib, OpenCV

Tools: Git, Docker, GCP, AWS, Linux, Power BI

EXPERIENCE

AI Engineer | [Showcase](#)

Jan. 2026 - Aug. 2026

City of Edmonton - Future Cities Institute

Edmonton, AB

- Developed CityVision, an end-to-end computer vision project funded by Mitacs, aiming to assist Edmonton's traffic management by analyzing volume and speed, saving **\$20,000** for the city.
- Improved UI responsiveness by migrating the app's tech stack from **R** to **React** and **Python**.
- Implemented traffic monitoring system using **OpenCV** and **RT-DETR**, leveraging **Apache Beam** and **GCP Dataflow** pipelines for scalable processing of **1,440 hours of video data** stored in **Google Cloud Storage**.
- Optimized computational resource efficiency by **33%** by implementing **background subtraction** module using frame differencing to omit parked cars from the study.
- Enhanced speed measurement accuracy by **27%** by calibrating intrinsic and extrinsic parameters to perform **3D reconstruction**, transforming perspectives into Bird's Eye View.
- Containerized the entire pipeline using **Docker** and deployed to **GCP Compute Engine**, ensuring environment consistency across development and production stages.

Database Developer

May. 2025 - Aug. 2025

Association of Korean Canadian Scientists and Engineers

Waterloo, ON

- Led the **team of 6** to develop the official AKCSE website using **Express** and **MongoDB**, improving accessibility to career and academic help resources for students.
- Optimized query efficiency and database scalability by **42%** through implementation of **dynamic CRUD**, while implementing **authentication** and access control features to strengthen data security.

PROJECTS

Computer Vision Football Analytics | *Python, OpenCV, Pandas* | [GitHub](#) | [Link](#)

- Developed a football analysis system using **YOLOv11** and fine-tuned with **10,000+** images to detect players, referees, and the ball, achieving **0.877** mAP50 with 35.7 ms per-frame latency.
- Implemented **K-means clustering** to automatically assign players to their team, according to their jersey colors.
- Enhanced players' metrics accuracy by **27%** by engineering **camera movement compensation** feature using **Optical Flow** to track metrics consistently even on camera moving environment.
- Deployed services on **AWS EC2** with **SQS**, and built a **Next.js** web app for video upload and real-time analysis.
- Developing AI agent layer using **LangChain** and **Gemini 1.5 Pro** to provide tactical recommendations.

Fullstack Academic Management System | *Spring, Javascript, HTML/CSS, MySQL, Bootstrap* | [GitHub](#)

- Built a full-stack academic management web app of **20+** pages with **50+** users using **Spring** and **MySQL**.
- Distributed project on **AWS EC2** with **Apache Tomcat**, with optimized database schema and RESTful APIs.
- Implemented **10+** essential features, **CRUD** operations of course postings and assignments, self implemented real-time chat, custom course search filters, and dynamic operations and views with the database.
- Reduced query response time by **30%** using nested SELECT statements to handle grouped ID increments.
- Ensured reliability with **200+** test cases and resolved issues such as multiple PK, time zone mismatches.

CERTIFICATIONS

- Machine Learning Specialization**, Stanford University | [Credential](#)
- Deep Learning Specialization**, DeepLearning.AI | [Credential](#)